

SSS 1-3

TEACHER PRACTICAL ACTIVITY MANUAL

Implementation guide for PA-401 to PA-500

DESIGN -> INSTRUMENT -> MODEL -> EVALUATE -> OPTIMIZE

Purpose	Turn the GrowMeal practical cards into safe, repeatable, age-appropriate teaching sessions.
Teacher role	Prepare the site, demonstrate technique, supervise safety, question learners, check evidence and connect the activity to the journal, mission cards and assessment system.
Core cycle	PREPARE -> DEMONSTRATE -> PRACTISE -> RECORD -> DISCUSS -> CONNECT -> CLEAN UP
Assessment	Use observation, student records and the linked practical/investigation evidence. Avoid grading recall alone.
Local adaptation	Substitute locally available materials when the learning objective, safety and measurement quality are preserved.

WHOLE-MANUAL SAFETY & IMPLEMENTATION STANDARD

Teachers must inspect the garden before each activity; establish boundaries; match tools to learner age; supervise water, cutting tools, electrical equipment, pumps, nutrient solutions and instruments; require handwashing after garden work; prohibit tasting unless a separate approved food-safety activity explicitly permits it; stop any activity when conditions are unsafe.

Before class	Check site, weather/working conditions, tools, labels, water, access routes and learner grouping.
During class	Demonstrate first; keep learners within assigned zones; monitor tool use and protect plants, structures and instruments.
After class	Count/clean/store tools; close water systems; restore labels; record damage, abnormal readings or incidents; wash hands.
Research integrity	Never invent, overwrite or silently delete observations. Flag questionable data and document repeat/reference checks.
Accessibility	Use peer roles, large-print/visual instructions, reachable observation stations and alternative recording methods where appropriate.

GARDEN 01 - Advanced Production Garden

Teaching focus: yield components, harvest index and resource-use efficiency.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Advanced Production Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-401 - FRAME - Advanced Production Garden

Learning objective	Guide students to frame a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-402 - DESIGN - Advanced Production Garden

Learning objective	Guide students to design a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-403 - INSTRUMENT - Advanced Production Garden

Learning objective	Guide students to instrument a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-404 - CALIBRATE - Advanced Production Garden

Learning objective	Guide students to calibrate a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-405 - MODEL - Advanced Production Garden

Learning objective	Guide students to model a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-406 - VALIDATE - Advanced Production Garden

Learning objective	Guide students to validate a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-407 - EVALUATE - Advanced Production Garden

Learning objective	Guide students to evaluate a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-408 - OPTIMIZE - Advanced Production Garden

Learning objective	Guide students to optimize a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-409 - STRESS-TEST - Advanced Production Garden

Learning objective	Guide students to stress-test a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-410 - RECOMMEND - Advanced Production Garden

Learning objective	Guide students to recommend a research or engineering investigation of yield components, harvest index and resource-use efficiency and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Production Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 02 - Crop Trial Garden

Teaching focus: replicated trials, blocks, variability and treatment effects.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Crop Trial Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-411 - FRAME - Crop Trial Garden

Learning objective	Guide students to frame a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-412 - DESIGN - Crop Trial Garden

Learning objective	Guide students to design a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-413 - INSTRUMENT - Crop Trial Garden

Learning objective	Guide students to instrument a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-414 - CALIBRATE - Crop Trial Garden

Learning objective	Guide students to calibrate a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-415 - MODEL - Crop Trial Garden

Learning objective	Guide students to model a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-416 - VALIDATE - Crop Trial Garden

Learning objective	Guide students to validate a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-417 - EVALUATE - Crop Trial Garden

Learning objective	Guide students to evaluate a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-418 - OPTIMIZE - Crop Trial Garden

Learning objective	Guide students to optimize a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-419 - STRESS-TEST - Crop Trial Garden

Learning objective	Guide students to stress-test a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-420 - RECOMMEND - Crop Trial Garden

Learning objective	Guide students to recommend a research or engineering investigation of replicated trials, blocks, variability and treatment effects and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Crop Trial Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 03 - Controlled Experiment Garden

Teaching focus: causal design, factorial effects, confounding and sensitivity.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Controlled Experiment Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-421 - FRAME - Controlled Experiment Garden

Learning objective	Guide students to frame a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-422 - DESIGN - Controlled Experiment Garden

Learning objective	Guide students to design a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-423 - INSTRUMENT - Controlled Experiment Garden

Learning objective	Guide students to instrument a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-424 - CALIBRATE - Controlled Experiment Garden

Learning objective	Guide students to calibrate a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-425 - MODEL - Controlled Experiment Garden

Learning objective	Guide students to model a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-426 - VALIDATE - Controlled Experiment Garden

Learning objective	Guide students to validate a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-427 - EVALUATE - Controlled Experiment Garden

Learning objective	Guide students to evaluate a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-428 - OPTIMIZE - Controlled Experiment Garden

Learning objective	Guide students to optimize a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-429 - STRESS-TEST - Controlled Experiment Garden

Learning objective	Guide students to stress-test a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-430 - RECOMMEND - Controlled Experiment Garden

Learning objective	Guide students to recommend a research or engineering investigation of causal design, factorial effects, confounding and sensitivity and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Controlled Experiment Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 04 - Irrigation Engineering Garden

Teaching focus: flow, head, pipe loss, uniformity and pump performance.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Irrigation Engineering Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-431 - FRAME - Irrigation Engineering Garden

Learning objective	Guide students to frame a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-432 - DESIGN - Irrigation Engineering Garden

Learning objective	Guide students to design a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-433 - INSTRUMENT - Irrigation Engineering Garden

Learning objective	Guide students to instrument a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-434 - CALIBRATE - Irrigation Engineering Garden

Learning objective	Guide students to calibrate a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-435 - MODEL - Irrigation Engineering Garden

Learning objective	Guide students to model a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-436 - VALIDATE - Irrigation Engineering Garden

Learning objective	Guide students to validate a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-437 - EVALUATE - Irrigation Engineering Garden

Learning objective	Guide students to evaluate a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-438 - OPTIMIZE - Irrigation Engineering Garden

Learning objective	Guide students to optimize a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-439 - STRESS-TEST - Irrigation Engineering Garden

Learning objective	Guide students to stress-test a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-440 - RECOMMEND - Irrigation Engineering Garden

Learning objective	Guide students to recommend a research or engineering investigation of flow, head, pipe loss, uniformity and pump performance and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Irrigation Engineering Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 05 - Water Productivity Garden

Teaching focus: water balance, ET and water-productivity indicators.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Water Productivity Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-441 - FRAME - Water Productivity Garden

Learning objective	Guide students to frame a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-442 - DESIGN - Water Productivity Garden

Learning objective	Guide students to design a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-443 - INSTRUMENT - Water Productivity Garden

Learning objective	Guide students to instrument a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-444 - CALIBRATE - Water Productivity Garden

Learning objective	Guide students to calibrate a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-445 - MODEL - Water Productivity Garden

Learning objective	Guide students to model a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-446 - VALIDATE - Water Productivity Garden

Learning objective	Guide students to validate a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-447 - EVALUATE - Water Productivity Garden

Learning objective	Guide students to evaluate a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-448 - OPTIMIZE - Water Productivity Garden

Learning objective	Guide students to optimize a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-449 - STRESS-TEST - Water Productivity Garden

Learning objective	Guide students to stress-test a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-450 - RECOMMEND - Water Productivity Garden

Learning objective	Guide students to recommend a research or engineering investigation of water balance, ET and water-productivity indicators and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Water Productivity Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 06 - Advanced Soilless Garden

Teaching focus: recirculating flow, pH/EC dynamics and residence time.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Advanced Soilless Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-451 - FRAME - Advanced Soilless Garden

Learning objective	Guide students to frame a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-452 - DESIGN - Advanced Soilless Garden

Learning objective	Guide students to design a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-453 - INSTRUMENT - Advanced Soilless Garden

Learning objective	Guide students to instrument a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-454 - CALIBRATE - Advanced Soilless Garden

Learning objective	Guide students to calibrate a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-455 - MODEL - Advanced Soilless Garden

Learning objective	Guide students to model a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-456 - VALIDATE - Advanced Soilless Garden

Learning objective	Guide students to validate a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-457 - EVALUATE - Advanced Soilless Garden

Learning objective	Guide students to evaluate a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-458 - OPTIMIZE - Advanced Soilless Garden

Learning objective	Guide students to optimize a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-459 - STRESS-TEST - Advanced Soilless Garden

Learning objective	Guide students to stress-test a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-460 - RECOMMEND - Advanced Soilless Garden

Learning objective	Guide students to recommend a research or engineering investigation of recirculating flow, pH/EC dynamics and residence time and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Advanced Soilless Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 07 - Sensor & Monitoring Garden

Teaching focus: accuracy, precision, calibration and monitoring-network design.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Sensor & Monitoring Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-461 - FRAME - Sensor & Monitoring Garden

Learning objective	Guide students to frame a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-462 - DESIGN - Sensor & Monitoring Garden

Learning objective	Guide students to design a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-463 - INSTRUMENT - Sensor & Monitoring Garden

Learning objective	Guide students to instrument a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-464 - CALIBRATE - Sensor & Monitoring Garden

Learning objective	Guide students to calibrate a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-465 - MODEL - Sensor & Monitoring Garden

Learning objective	Guide students to model a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-466 - VALIDATE - Sensor & Monitoring Garden

Learning objective	Guide students to validate a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-467 - EVALUATE - Sensor & Monitoring Garden

Learning objective	Guide students to evaluate a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-468 - OPTIMIZE - Sensor & Monitoring Garden

Learning objective	Guide students to optimize a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-469 - STRESS-TEST - Sensor & Monitoring Garden

Learning objective	Guide students to stress-test a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-470 - RECOMMEND - Sensor & Monitoring Garden

Learning objective	Guide students to recommend a research or engineering investigation of accuracy, precision, calibration and monitoring-network design and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Sensor & Monitoring Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 08 - Smart Irrigation Demonstrator

Teaching focus: thresholds, hysteresis, feedback and adaptive control.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Smart Irrigation Demonstrator.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-471 - FRAME - Smart Irrigation Demonstrator

Learning objective	Guide students to frame a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-472 - DESIGN - Smart Irrigation Demonstrator

Learning objective	Guide students to design a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-473 - INSTRUMENT - Smart Irrigation Demonstrator

Learning objective	Guide students to instrument a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-474 - CALIBRATE - Smart Irrigation Demonstrator

Learning objective	Guide students to calibrate a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-475 - MODEL - Smart Irrigation Demonstrator

Learning objective	Guide students to model a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-476 - VALIDATE - Smart Irrigation Demonstrator

Learning objective	Guide students to validate a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-477 - EVALUATE - Smart Irrigation Demonstrator

Learning objective	Guide students to evaluate a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-478 - OPTIMIZE - Smart Irrigation Demonstrator

Learning objective	Guide students to optimize a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-479 - STRESS-TEST - Smart Irrigation Demonstrator

Learning objective	Guide students to stress-test a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-480 - RECOMMEND - Smart Irrigation Demonstrator

Learning objective	Guide students to recommend a research or engineering investigation of thresholds, hysteresis, feedback and adaptive control and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Smart Irrigation Demonstrator.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 09 - Research & Design Garden

Teaching focus: requirements, concept selection, prototyping and testing.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Research & Design Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-481 - FRAME - Research & Design Garden

Learning objective	Guide students to frame a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-482 - DESIGN - Research & Design Garden

Learning objective	Guide students to design a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-483 - INSTRUMENT - Research & Design Garden

Learning objective	Guide students to instrument a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-484 - CALIBRATE - Research & Design Garden

Learning objective	Guide students to calibrate a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-485 - MODEL - Research & Design Garden

Learning objective	Guide students to model a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-486 - VALIDATE - Research & Design Garden

Learning objective	Guide students to validate a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-487 - EVALUATE - Research & Design Garden

Learning objective	Guide students to evaluate a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-488 - OPTIMIZE - Research & Design Garden

Learning objective	Guide students to optimize a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-489 - STRESS-TEST - Research & Design Garden

Learning objective	Guide students to stress-test a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-490 - RECOMMEND - Research & Design Garden

Learning objective	Guide students to recommend a research or engineering investigation of requirements, concept selection, prototyping and testing and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Research & Design Garden.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 10 - Integrated Food Systems Laboratory

Teaching focus: system boundaries, balances, mismatches, cascades and scenarios.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Integrated Food Systems Laboratory.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-491 - FRAME - Integrated Food Systems Laboratory

Learning objective	Guide students to frame a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students frame according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-492 - DESIGN - Integrated Food Systems Laboratory

Learning objective	Guide students to design a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students design according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-493 - INSTRUMENT - Integrated Food Systems Laboratory

Learning objective	Guide students to instrument a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students instrument according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-494 - CALIBRATE - Integrated Food Systems Laboratory

Learning objective	Guide students to calibrate a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students calibrate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-495 - MODEL - Integrated Food Systems Laboratory

Learning objective	Guide students to model a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students model according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-496 - VALIDATE - Integrated Food Systems Laboratory

Learning objective	Guide students to validate a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students validate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-497 - EVALUATE - Integrated Food Systems Laboratory

Learning objective	Guide students to evaluate a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students evaluate according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-498 - OPTIMIZE - Integrated Food Systems Laboratory

Learning objective	Guide students to optimize a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students optimize according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-499 - STRESS-TEST - Integrated Food Systems Laboratory

Learning objective	Guide students to stress-test a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students stress-test according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-500 - RECOMMEND - Integrated Food Systems Laboratory

Learning objective	Guide students to recommend a research or engineering investigation of system boundaries, balances, mismatches, cascades and scenarios and produce evidence suitable for design, optimization or operational decisions.
Materials	Research journal, appropriate instruments/sensors, calibration/reference materials where approved, calculator/computer where available, system drawings and safety equipment.
DIY / low-cost option	Use low-cost reference measurements, locally fabricated sampling frames, marked vessels and paper-based models where specialist hardware is unavailable.
Teacher preparation	Specify system boundary, performance metric, assumptions, constraints, test protocol, QC/validation method and safety controls for the Integrated Food Systems Laboratory.
Step-by-step delivery	1. Frame the research/design objective. 2. Define metrics, constraints and acceptance criteria. 3. Establish instrument IDs and QC. 4. Students recommend according to protocol. 5. Process data/model scenarios. 6. Validate or compare with reference evidence. 7. Evaluate uncertainty and trade-offs. 8. Recommend optimization and a next-stage test.
Questions to ask	What decision must the evidence support? Which assumptions matter most? How was measurement quality checked? Does validation support the model/design? What trade-off constrains optimization?
Expected evidence	Protocol, traceable dataset, QC/calibration note, calculation/model, validation evidence, uncertainty statement and technically defensible recommendation.
Differentiation	Provide structured protocols and calculation templates; extension students conduct scenario, sensitivity or uncertainty analysis.
Troubleshooting	If instrumentation or model output is unreliable, suspend decision-making, diagnose the data-quality failure, establish a reference check and document corrective action.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

LEVEL IMPLEMENTATION REVIEW

Use this page after completing the level's practical sequence.

Strongest learner competency	
Competency needing reteaching	
Garden/system needing maintenance	
Materials to replenish	
Safety or accessibility improvement	
Next mission/assessment priority	